

The SignalFire State of Tech Talent Report - 2025

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Tech's talent gap widens: new grad hiring plummets, Texas loses startup talent, and Anthropic's 80% retention dominance.

Over the last couple of years, we've seen companies rewrite the playbook for hiring AI and tech talent. In a landscape defined by fierce competition and rapid change, our latest tech talent report zeroes in on the bold moves, early trends, and strategic pivots reshaping how companies attract and retain top-tier talent in 2025 and beyond.

At SignalFire, our Beacon AI platform tracks over 650 million professionals and 80 million organizations to give us a front-row seat to the talent transformations reshaping the industry. Our latest report reveals some sharp shifts:

- Entry-level hiring is collapsing
- A generational hiring shift is leaving new graduates behind
- Elite AI labs are locking in top talent, and retaining them
- Geographic power centers are evolving fast

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analyses that inform SignalFire's investment strategy and enable portfolio company success, and builds tools and systems that guide decision-making across the firm. Before SignalFire, Asher was a Staff Data Scientist at Block's CashApp, where he was responsible for analytics for banking products including credit offerings and direct deposits. Prior to that, he was a Data Scientist at Medium, where he designed experimentation tools and the data infrastructure underlying Medium's writer partner programs.

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While headlines continue to highlight labor market shifts amidst political change and intense AI rivalry among tech giants, our data reveals a more nuanced story. This isn't just about layoffs or remote work. It's about a fundamental reset in how, where, and who companies are hiring to build the next generation of technology.

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#1 Tech's lost generation? New grad hiring drops 50% compared to pre-pandemic levels

The tech world has long been synonymous with innovation, breakneck growth, and boundless opportunities. The door to tech once swung wide open for new grads. Today, it's barely cracked. The industry's obsession with hiring bright-eyed grads right out of college is colliding with new realities: smaller funding rounds, shrinking teams, fewer new grad programs, and the rise of AI.

Everyone took a hit in 2023, but while hiring bounced back in 2024 for mid- and senior-level roles, the cut keeps getting deeper for new grads:

- **Big Tech:** New grads now account for just 7% of hires, with new hires down

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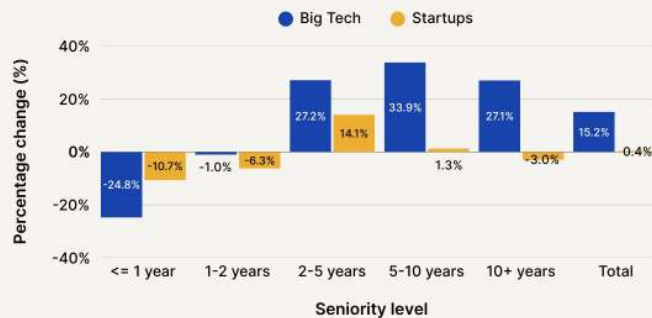
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25% from 2023 and over 50% from pre-pandemic levels in 2019.

- **Startups:** New grads make up under 6% of hires, with new hires down 11% from 2023 and over 30% from pre-pandemic levels in 2019.

New grads left behind: Entry-level tech hiring fell 25% in 2024 despite market recovery

Percentage change in number of new hires by seniority (2023-2024)

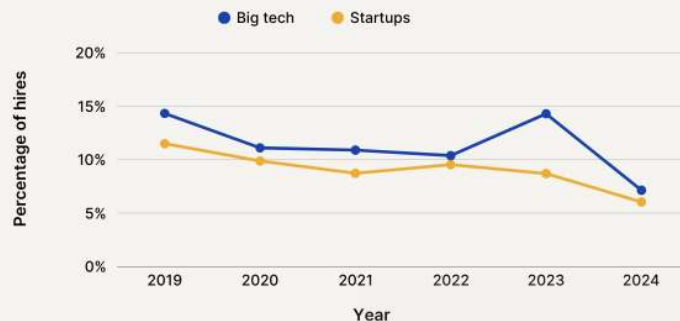


*Big tech represents the top 15 technology companies by market cap.
*Startups represents companies funded by the Top 100+ VC firms that closed a Seed through Series C round in the previous 4 years.

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Gen Z squeeze: Entry-level share of hires down 50% from pre-pandemic levels

Percentage of new hires with one year or less of total experience at time of hire.



*Big tech represents the top 15 technology companies by market cap.
*Startups represents companies funded by the top 100+ VC firms that closed a Seed through Series C round in the previous 4 years.

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As budgets tighten and AI capabilities increase, companies are reducing their investment in new grad opportunities. The latest data from the Federal Reserve Bank of New York shows that the unemployment rate

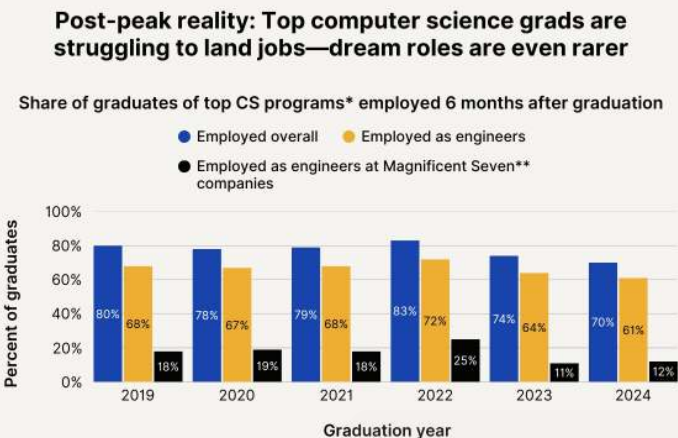
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for new college grads has risen 30% since bottoming out in September 2022, versus about 18% for all workers. The perception gap isn't helping—55% of employers say Gen Z struggles with teamwork, and 37% of managers said they'd rather use AI than hire a Gen Z employee.

Even top computer science grads aren't spared. As demand for junior roles declines, even highly credentialed engineering grads are struggling to break into tech, especially at the Big Tech companies. The share of new graduates landing roles at the Magnificent Seven (Alphabet, Amazon, Apple, Meta, Microsoft, NVIDIA and Tesla) has dropped by more than half since 2022.



The experience paradox

It's not just a hiring slowdown, it's a shift in expectations. Today's tech employers aren't looking for potential, they're looking for

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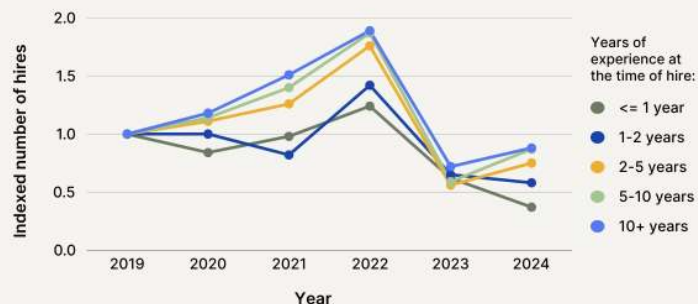
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proof. That's left new grads stuck in a Catch-22: you need experience to get the job, but you need the job to get experience.

In a world of leaner teams and tighter budgets, there's little room for investment in training. As the pipeline for new talent shrinks at breakneck speed, it's creating fierce competition for the few entry-level roles that remain. The cruel irony? Companies are posting junior roles but filling them with senior individual contributors (ICs) —a phenomenon known as the experience paradox.

Big Tech: 2023 hiring crash hits all levels 2024 rebound favors seniors, while new grads get left behind

Indexed new hire count from 2019-2024 shows Big Tech slashed early-career hires, but cautiously rehired mid-career employees and senior execs.



*This graph only includes employees of the top 15 technology companies by market cap.

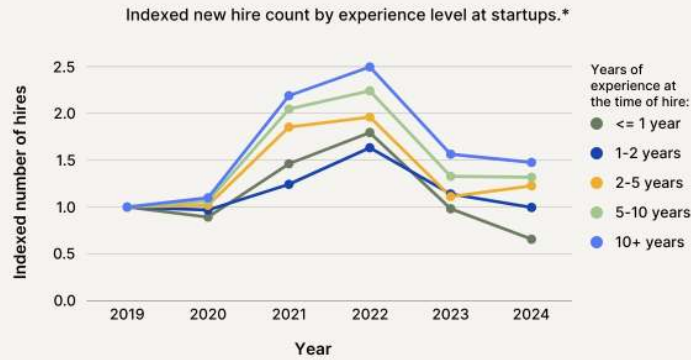
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Startups: Job market steadies, but early-career talent keeps taking the hit



*This graph only includes employees at startups funded by the top 100+ VC firms that closed a Seed through Series C round in the previous 4 years.

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Series A squeeze or AI hype: What's really stalling Gen Z hiring?

AI gets a lot of blame for wiping out junior roles post-2022, but the real story is more nuanced, as you can see from the 200+ comments on this [Hacker News post](#) and this recent article from [The Atlantic](#). Yes, automation is already [replacing some routine tasks](#), but the bigger driver may be the end of the “free money madness” driven by low interest rates that we saw in 2020-2022, along with the overhiring and inflation it led to. Now, with tighter budgets and shorter runways, companies are hiring leaner and later. [Carta data](#) shows that Series A tech startups are 20% smaller than they were in 2020.

This shift isn't just about hiring less—it's a hiring reset. As AI tools [take over more](#) routine, entry-level tasks, companies are

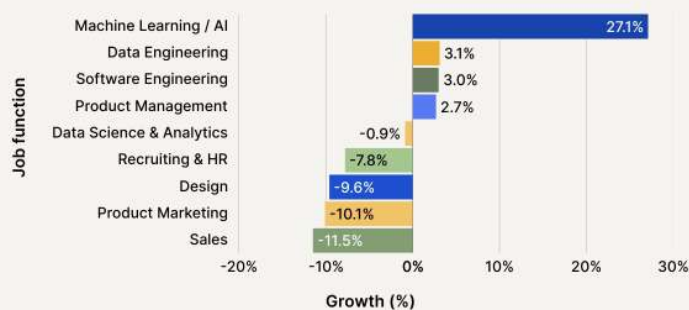
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prioritizing roles that deliver high-leverage technical output. Big Tech is doubling down on machine learning and data engineering, while non-technical functions like recruiting, product, and sales keep shrinking, making it especially tough for Gen Z and early-career talent to break in.

AI surge reshapes hiring: Big Tech prioritizes technical roles, slows GTM hiring

2023 vs. 2024: Big Tech hiring change by job function



*Big tech represents the top 15 technology companies by market cap.

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But despite the hype, AI hasn't wiped out entire job categories—yet. So far, the fallout has hit new grads hardest, while demand for experienced engineers is still rising. As AI capabilities continue to grow, the real differentiator will be talent, **and no one's playing the game harder than the top AI labs.**

#2 AI Labs: Anthropic sets the pace in the talent race

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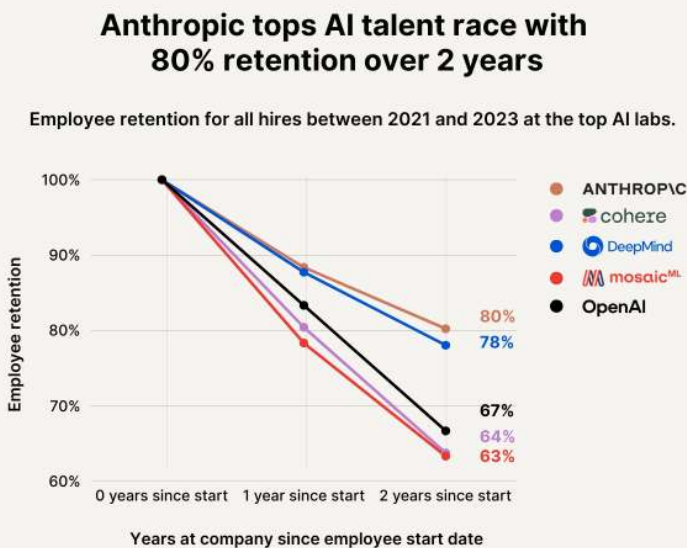
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November 2022 didn't just mark the launch of ChatGPT—it kicked off the AI talent race.

We analyzed retention across top AI labs, and one clear leader emerged: Anthropic.

An impressive 80% of employees hired at least 2 years ago at Anthropic were still at the company at the end of their second year—a figure that stands out in an industry known for high turnover. DeepMind follows closely at 78%, while OpenAI's retention trails at 67% but remains on par with large FAANG companies like Meta (64%).



*This graph represents retention over the last two years of employees who started working at the top AI labs in 2021 or later. It excludes employees hired after Feb. 2023.

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With outsized salary packages at DeepMind and even Senate scrutiny over Big Tech talent poaching, the AI talent war is real, and the spoils go to those who can attract and keep the best minds.

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Anthropic's winning formula and where they're hiring from:

Anthropic's 80% retention isn't just impressive—it's a strategic advantage. In a field where turnover means lost time and money, keeping top talent consistently is a game-changer. And it's not just about keeping talent, it's about winning it.

- **Talent poaching with precision:**

Anthropic is siphoning top talent from two of its biggest rivals: OpenAI and DeepMind. Engineers are **8 times more likely** to leave OpenAI for Anthropic than the reverse. From DeepMind, the ratio is nearly **11:1** in Anthropic's favor. Some of that's expected—Anthropic is the hot new startup, while DeepMind's larger, tenured team is ripe for movement. But the scale of the shift is striking.

- **Beyond the salary:** While Big Tech leans on big paychecks and brand cachet, Anthropic's edge is its unique culture that embraces unconventional thinkers and gives employees true autonomy to drive impact. Flexible work options, no title politics or forced management tracks—just clear paths for career growth. Employees say Anthropic embraces intellectual discourse and researcher autonomy, making it a magnet for AI talent stifled by bureaucracy elsewhere.

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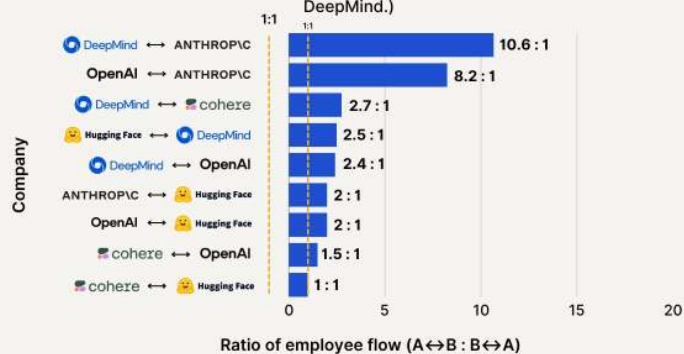
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- **Claude is quickly emerging as a favorite among developers** by several informal measures, and that kind of affinity can influence career decisions. Engineers often gravitate toward companies whose products they admire and use, and that perceived product resonance may be giving Anthropic an edge in recruiting.

AI's talent shuffle: Mapping the movement between top AI labs

Bidirectional movement between major AI labs. (For example, for every 10.6 DeepMind employees that moved to Anthropic, 1 Anthropic employee moved to DeepMind.)



*This graph shows the ratios of bidirectional movement between the top AI labs.
 *We excluded newer AI labs (e.g., DeepSeek and xAI/Grok) from our analysis because they were not in operation for the whole of 2024.

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It's not just rival AI labs feeling the pull—**Big Tech has become a prime hunting ground, too.** Google, Meta, Microsoft, Amazon, and Stripe serve as primary talent pools for AI labs, and Anthropic has been particularly successful at poaching senior researchers and engineers from these companies.

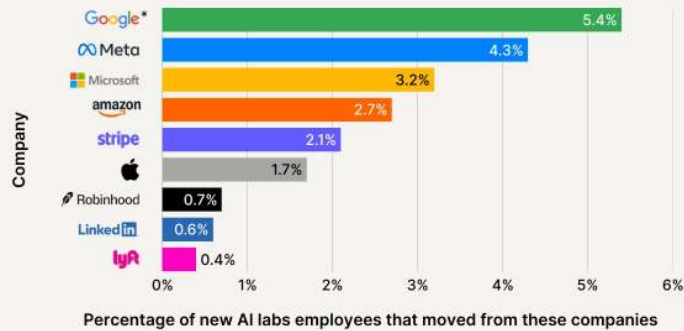
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Which tech giants are losing their best talent to AI labs?

AI labs are poaching top talent from Big Tech, with Google, Meta, Amazon, and Microsoft as prime targets.



*Google (not including DeepMind)

*This graph shows the previous employers of AI lab joiners in 2024 (percentage of total).

*Exclusions: We excluded some newer AI labs as we didn't have complete data for 2024.

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#3 Hot spots, cooling zones, and America's new tech hubs

The tech talent map continues to shift. While San Francisco and New York remain dominant, rising hubs like Toronto, Miami, and San Diego are gaining ground. Meanwhile, Texas cities like Austin are losing momentum, prompting founders to rethink hiring and compensation strategies. More companies are embracing hub-and-spoke models and tailoring compensation philosophies to ensure they secure the right talent mix across diverse locations.

Key trends shaping the new geographic playbook

Hiring is slowing in most metros, but we're seeing a reconsolidation of tech jobs around major tech hubs like the Bay Area, Seattle,

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and NYC. Here are the key geographic trends we saw in 2024:

1. Silicon Valley and New York City hold strong:

- **High talent, high cost:** SF and NYC still anchor the AI ecosystem, with over 65% of AI engineers based in these two metros. Despite rising housing costs, shrinking salary advantages, and remote work flexibility, SF and NY continue to attract more tech talent.

2. The sunshine surge - Miami and San Diego are rising fast:

These two cities are attracting tech talent not with massive budgets, but with sun, lifestyle, and lower living costs.

- **Miami's mix** of tax perks and quality of life has fueled a 12% jump in AI roles, according to recent hiring data.

- **San Diego** saw a 7% rise in Big Tech roles, even as startups in the region lost 3.5% of their workforce in 2024, suggesting talent is being poached upward. In 2024, San Diego County startups raised \$5.7 billion in venture capital, marking one of the region's best-performing years on record.

3. Texas cools off: Are Austin and Houston losing their luster?

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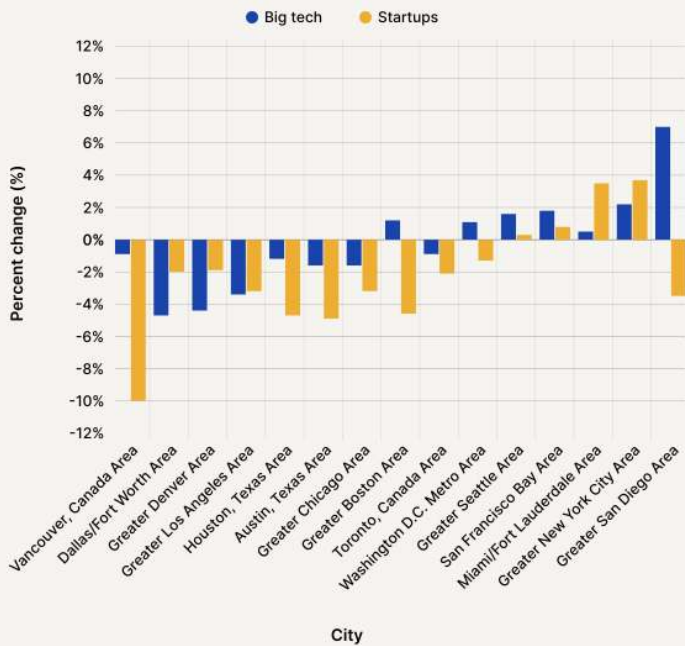
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Once the darlings of tech growth, Austin and Houston have been losing startup talent. Lagging infrastructure, a cultural mismatch, fluctuating housing costs, and a renewed emphasis on hybrid RTO policies are motivating startup employees to live closer to traditional hubs. Last year, Austin, which was a post-pandemic growth leader, saw a 6% drop in headcount at VC-backed startups. Houston's drop is even steeper at 10.9%.

Big hubs gain jobs as startup team size shrinks; Miami and San Diego defy this trend

YoY growth in big tech and startup employment in (2024 vs. 2023)



*This graph shows the year-over-year percentage growth in tech jobs in 2024 relative to the previous year, in North America (excludes contractors and gig workers)
*We only included startups funded by the top 100+ VC firms that closed a Seed through Series C round in the previous 4 years.



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This isn't just a story about regional reshuffling—it's a signal of something bigger. The media's return-to-office narrative is overstated, and companies are rethinking what really matters. The new model? **Proximity over presence.** For many tech

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companies, it's not about clocking in five days a week—it's about being close enough for hybrid schedules and anchor days. The result: a surge in in-state hiring as employers strike a new balance between flexibility and face time.

#4 Hits and misses from 2024 and predictions for 2025

Some of our predictions from last year's report held strong while others veered off course. Here's what we got right—and what we predict is next.

2024: What we saw coming:

- **Fractional work stuck around:** Fractional roles are thriving—even at the top. CMOs, CFOs, and CTOs are increasingly working as consultants. Whether this trend holds in a stronger market remains to be seen, but for now, this model seems to be working for companies and execs alike.
- **Cybersecurity kept growing:** AI-powered threats are on the rise, and the demand for cybersecurity talent has continued to grow as predicted. Compensation is up, roles are harder to fill, and hiring urgency is higher than ever.

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- **Remote work didn't die—it evolved:** The RTO debate rages on, but the reality is more nuanced. Companies are increasingly adopting hybrid models while talent continues to demand flexibility. We won't know the long-term balance until the next economic cycle resets the supply-demand equation.

Our predictions for 2025:

- **The rise of the generalist engineer:** Specialists won the last decade. Generalists may win the next. With tools like Copilot, Replit, and Cursor maturing, engineers don't need deep ML expertise to build with AI. Companies will prioritize flexible, collaborative, generalist engineers who can move fast and collaborate effectively with powerful tools—no PhD required.
- **The year of the equity advisor:** With lean startups hesitant to hire for both junior and C-suite roles, founders will tap seasoned experts as equity advisors. Carta data shows these roles are more affordable than before, offering startups a low-cost way to gain experience and mentorship without inflating their burn rate.
- **New jobs—not just disappearing ones:** While headlines warn of jobs lost to AI, we see another shift: *new* roles emerging. Expect to see titles like AI

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governance lead, AI ethics and privacy specialists, agentic AI engineers, and non-human security ops specialists become commonplace. It'll take time to scale, but these are some of the roles new grads should be paying attention to.

People, not just technology, will shape 2025 and beyond

The past year made one thing clear: technology alone doesn't build the future, people do. Anthropic's retention edge and the reshaping of talent hubs prove that the real advantage lies in how you hire, grow, and keep great talent.

What it means for the road ahead:

- **For new grads:** The training wheels are gone. With fewer entry-level roles, the path forward will rely on bootcamps, open-source, freelancing, and creative projects. It's not enough to just master the latest AI tools; learn to fix their flaws—debugging messy, machine-generated code may be your superpower.

- **For employers:** AI might reduce the short-term need for junior hires, but skipping them entirely risks breaking the long-term talent

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pipeline. The industry's future depends on equipping the next generation with skills that grow alongside the evolving technology landscape.

(Read our article on how to build a multi-generational company [here](#).)

Methodology Note:

This report uses data from our proprietary Beacon AI platform, an intelligence engine that tracks 650+ million individuals and 80+ million organizations. We analyze millions of data points on hiring trends, geographic movements, and more to spot emerging talent and help our portfolio companies build teams and products faster.

Here's the approach we used to analyze the data for this report:

- "Big tech" represents the top 15 technology companies by market cap.

- "Startups" represents companies funded by the Top 100 VC firms that closed a Seed through Series C round in the previous 4 years.

- "Top computer science graduates" represents graduates from the top 20 engineering programs in the U.S. according to the U.S. News' Best Undergraduate Engineering Programs Rankings.

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- For the AI Labs retention analysis, we excluded some newer AI labs (like DeepSeek and xAI/Grok) because they had not been operating over the whole time period we looked at (2023-2024). We focused on entities with distinct LinkedIn company profiles to ensure clean, accurate employee data. Meta AI wasn't included, as it's not listed as a distinct entity from Meta as a whole on LinkedIn.

- All employee retention numbers are derived from public, self-reported LinkedIn profiles, which may include inaccuracies or incomplete employment date ranges.

- "Magnificent Seven" companies (MAG7) include Alphabet, Amazon, Apple, Meta Platforms, Microsoft, NVIDIA, and Tesla.

About SignalFire

SignalFire is the first VC firm built like a technology company to better serve the needs of founders as they build and scale their startups. With approximately \$3B in assets under management, SignalFire invests in applied AI companies from pre-seed to Series B in key sectors, including healthcare, cybersecurity, infrastructure, consumer, and other enterprise verticals.

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The firm's Beacon AI platform tracks over 650M employees and 80M organizations, giving the firm an unmatched data advantage in identifying and supporting world-class startups. Its sector-focused investors and a dedicated team of seasoned operators drive SignalFire at startup speed. They provide support across a company's full lifecycle through data and resources tailored by growth stage, plus a diverse ecosystem of partners and customers. Notable investments include Frame.io, Grammarly, Grow Therapy, EvenUp, and Stamp.li.

Learn more at www.signalfire.com.

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